A PROJECT REPORT

on

"Genetic Algorithm-Optimized Support Vector Machine for Pediatric Pneumonia Detection from Chest X-Ray Images"

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BACHELOR'S DEGREE IN

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

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ABSTRACT

Pneumonia is a serious lung infection that kills thousands of children every year, especially in regions where doctors and radiologists are hard to find. Diagnosing it from chest X-rays usually needs trained specialists, which is not always possible in smaller clinics. This project tries to address that gap by building an automated detection system using classical machine learning — no deep learning, no GPU required.

We implemented a GA-SVM model in Python using Scikit-Learn, Scikit-Image, and the DEAP library, running on Google Colab. Each chest X-ray was converted to grayscale, resized to 64×64 pixels, and then described using HOG and LBP features — giving a 1,774-dimensional vector per image. Instead of manually tuning the SVM, we used a Genetic Algorithm (population = 20, generations = 15) to find the best values for C and γ automatically. The dataset we used is the publicly available Kaggle Chest X-Ray Images dataset [2], which contains 5,856 pediatric X-rays labelled as NORMAL or PNEUMONIA.

On the held-out test set of 1,172 images, our model got 94.80% accuracy, 98% recall for pneumonia cases, and an AUC of 0.9815. Importantly, the whole inference runs in under one second on CPU — so it could actually be used in a clinic without any special hardware. One limitation worth noting is that all images came from a single hospital in Guangzhou, which may mean the model needs further testing before it works reliably in other settings.

Keywords: Pediatric Pneumonia, Chest X-Ray, Support Vector Machine, Genetic Algorithm, HOG, LBP, Computer-Aided Diagnosis, Medical Image Classification.

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Chapter 1

Introduction

1.1 What is Pneumonia?

Pneumonia is an infection that inflames the air sacs (alveoli) in one or both lungs. When someone gets it, those sacs fill up with fluid or pus, which is why breathing becomes so hard. It can be caused by bacteria like Streptococcus pneumoniae, viruses like influenza or RSV, and sometimes fungi. Doctors usually confirm the diagnosis with a chest X-ray because it shows the fluid buildup clearly, and it is cheaper and faster than most other tests.

Different types exist — Community-Acquired Pneumonia (CAP) is what people typically catch outside hospitals, while Hospital-Acquired Pneumonia (HAP) is more serious because those bacteria tend to be drug-resistant. In immunocompromised patients, even normally harmless organisms can cause Opportunistic Pneumonia.

1.2 What is Pediatric Pneumonia?

Pediatric pneumonia is simply pneumonia in children — but it behaves quite differently from the adult version. Children under five have weaker immune systems, smaller lung fields, and their X-rays are harder to read because the rib angles and tissue density keep changing as they grow. According to WHO [7], pneumonia alone accounts for about 14% of all deaths in children under five globally, which made it one of the reasons we chose this problem.

The signs in young children — tachypnoea, nasal flaring, chest in-drawing — are different from what adults show, and clinicians without pediatric training can easily miss them. Automated detection can serve as a second opinion in those cases.

1.3 Target Age Range

Our project focuses on children aged 1 to 5 years. This matches the dataset we used — the Kaggle Chest X-Ray Images dataset by Kermany et al. [2], which was collected at the Guangzhou Women and Children's Medical Center and covers this exact cohort. This age group also carries the highest pneumonia mortality burden globally, so it made sense to start here.

1.4 Why is Early Detection Needed?

The short answer is that waiting too long can be fatal. Children under five can deteriorate from mild symptoms to respiratory failure within hours. Bacterial pneumonia, if caught early, responds well to standard antibiotics — but the same treatment does almost nothing once sepsis sets in. Three reasons stood out to us when we were reading through the literature:

1. Speed of deterioration: A child who looks slightly unwell in the morning can be in respiratory failure by evening. Early flagging changes the treatment window completely.
2. Antibiotic stewardship: If you do not know whether it is bacterial or viral, you risk either under-treating or over-prescribing antibiotics. An accurate diagnosis helps avoid both.
3. Non-specific symptoms: Fever, cough, and poor feeding are common to many childhood illnesses. Without imaging support, a clinical-only diagnosis has a known false-negative rate.

1.5 Why is Automated Detection Essential?

WHO reports [7] that roughly 740,000 children under five died from pneumonia in 2019 — the vast majority in low- and middle-income countries (LMICs). The radiologist-to-population ratio in parts of Sub-Saharan Africa is as low as 1 per 100,000 people. Chest X-ray machines exist in many district hospitals, but trained readers often do not.

In our system, after a chest X-ray is uploaded, the detection pipeline processes it in approximately 0.8 seconds on CPU hardware using the pre-extracted HOG and LBP features and the trained SVM. That kind of speed is actually useful in a real triage setting. We are not claiming it replaces a radiologist — but in a clinic where none is available, a system that flags likely pneumonia cases with 98% recall is meaningfully better than nothing.

We deliberately avoided deep learning for this reason: a CNN requires a GPU at inference time, costs thousands of dollars in hardware, and needs IT infrastructure to maintain. Our GA-SVM runs on any standard computer. That trade-off in accuracy (94.8% vs deep learning's 93–96%) seemed worth it for practical deployability.

Chapter 2
Literature Review and Basic Concepts

2.1 Overview of Prior Work

Table 2.1 presents a comprehensive review of prior research on automated pneumonia and chest disease detection. The table includes all major algorithms, datasets, and limitations identified in prior work.

Table 2.1: Literature Review – Automated Pneumonia Detection

Authors / Year	Method	Acc.	Limitation
Rajpurkar et al. (2017)	DenseNet-121 (CheXNet)	90.3%	Not pediatric, GPU only
Kermany et al. (2018)	Inception V3, Transfer Learning	92.8%	No hyperparameter opt.
Stephen et al. (2019)	Custom 7-layer CNN	93.7%	Black-box, no interpretability
Islam et al. (2020)	CNN + Random Forest	87.0%	Random feature selection
Liang & Zheng (2020)	VGG-16 + SVM	88.5%	Non-public dataset
Chouhan et al. (2020)	Ensemble of 5 CNNs	96.4%	Very high compute, not LMIC-ready
Kumar et al. (2022)	HOG + SVM (grid search)	89.4%	Manual grid search, no GA
Zhang et al. (2023)	Attention-based CNN	94.1%	High compute, GPU required
Proposed (2026)	**GA-SVM + HOG + LBP**	**94.8%**	**CPU-ready, leakage-free eval.**

2.2 Algorithms Used in Previous Work

CNNs have dominated medical image classification since 2017, particularly after Rajpurkar et al. introduced CheXNet [1] — a 121-layer DenseNet that reportedly matched radiologist-level performance on pneumonia detection. After that, most published work went the deep learning route. Kermany et al. [2] used Inception V3 with transfer learning and got 92.8% accuracy. Stephen et al. [6] built a 7-layer custom CNN and reached 93.7%, while Chouhan et al. [5] combined five CNN architectures in an ensemble and pushed accuracy up to 96.4%. The pattern is clear: the more compute you throw at it, the better the number — but none of these systems can run without a GPU.

The classical ML side is less crowded. Kumar et al. [12] used HOG features with an SVM tuned by grid search and got 89.4%. That is the closest prior work to ours. The difference in our approach is that we replaced grid search with a GA, added LBP features alongside HOG, and enforced a strict leakage-free evaluation. Those three changes together account for the 5.4% accuracy improvement we observed over their baseline. Islam et al. [4] combined CNN features with a Random Forest classifier at 87.0%, and Liang and Zheng [3] used VGG-16 as a feature extractor with an SVM classifier at 88.5%, though on a non-public dataset, so direct comparison is difficult. No prior work we found combines GA-based hyperparameter search with HOG+LBP features under a proper leakage-free train/val/test protocol — that is the gap this project addresses.

2.3 Core Technical Concepts

2.3.1 Support Vector Machine (SVM)

SVM finds the optimal separating hyperplane that maximizes the margin between classes. For non-linearly separable data, the RBF kernel maps data to a higher-dimensional space: $K(Xi, Xj) = \exp(-\gamma\|Xi - Xj\|^2)$. The regularisation parameter C controls the trade-off between margin maximization and misclassification penalty.

2.3.2 Genetic Algorithm (GA)

A Genetic Algorithm is an optimization method inspired by biological evolution — it maintains a population of candidate solutions and iteratively improves them through selection, crossover, and mutation. In our implementation using the DEAP library [14], each individual in the population encodes three SVM hyperparameters: $\log_{10}(C)$ in the range [−2, 4], $\log_{10}(\gamma)$ in [−5, 1], and a kernel index (0=RBF, 1=Linear, 2=Poly). We used logarithmic encoding for C and γ because their effects on SVM performance span multiple orders of magnitude.

Tournament selection with size 3 picks parents for crossover — three individuals are sampled randomly, and the best-performing one is chosen. We used blend crossover (α=0.3) rather than simple two-point crossover because it works better on continuous-valued genes like log(C). Gaussian mutation (σ=0.5, indpb=0.4) then perturbs individual genes with 40% probability per gene. Fitness is just the validation accuracy of the SVM trained with that individual's decoded parameters. The whole thing ran for 15 generations and converged by generation 9, which is earlier than we expected — it means the search space was not too rugged for the population to navigate.

2.3.3 HOG Features

Histogram of Oriented Gradients (HOG) divides the image into 8×8 pixel cells, computes gradient orientation histograms with 9 bins, and normalizes over 2×2 cell blocks. For a 64×64 image, this yields 1,764 features capturing edges, boundaries, and structural patterns in lung fields.

2.3.4 LBP Features

Local Binary Pattern (LBP) encodes each pixel by comparing it to P equally-spaced neighbours on a circle of radius R. With P=8, R=1, uniform mapping, and a 10-bin histogram, LBP adds 10 texture features per image, capturing alveolar texture and parenchymal density changes.

Chapter 3
Problem Statement and Requirement Specifications

3.1 Problem Statement

Despite chest X-ray machines being widely available in developing-world hospitals, the expertise to interpret them is absent in rural and underserved regions. The result is diagnostic delay and preventable child deaths from pneumonia. Manual radiologist review is expensive, slow, and unavailable round-the-clock. Existing deep learning CAD systems require GPU hardware costing thousands of dollars, making them impractical for primary-care clinics.

This project addresses the following specific problem: "Design and implement a computationally efficient, automated binary classification system that can accept a pediatric chest X-ray image as input and accurately predict whether the child has pneumonia (PNEUMONIA) or not (NORMAL), using only CPU-compatible machine learning methods, with a validated accuracy exceeding 90% and pneumonia recall exceeding 95%."

3.2 Project Planning / Timeline

Table 3.1: Project Development Timeline

Phase	Activity	Duration	Responsibility
1	Literature review & dataset acquisition	2 weeks	All members
2	Preprocessing & feature extraction	2 weeks	Ayush, Ankit
3	GA implementation & SVM training	3 weeks	Aryan, Tanisha
4	Evaluation & result analysis	2 weeks	Tannistha
5	Report writing & presentation	1 week	All members

3.3 Software Requirements Specification (SRS)

3.3.1 Functional Requirements

Table 3.2: Functional Requirements

ID	Requirement
FR-1	The system shall accept a JPEG or PNG chest X-ray image as input
FR-2	The system shall extract HOG and LBP features from the input image
FR-3	The system shall classify the image as NORMAL or PNEUMONIA
FR-4	The system shall output a confidence percentage for each class
FR-5	The system shall achieve ≥90% accuracy on the held-out test partition
FR-6	The system shall complete inference in <2 seconds on CPU hardware

3.3.2 Non-Functional Requirements

Reproducibility: SEED=42 set for all random operations. Portability: No GPU required at inference time. Leakage-free: Feature scaler fitted only on training data. Interpretability: Model outputs probability scores for both classes.

3.4 System Design

The system follows a five-stage pipeline: (1) Image preprocessing – grayscale conversion and resize to 64×64 pixels; (2) Feature extraction – HOG (1,764-d) and LBP (10-d) concatenated to 1,774-d vector; (3) Leakage-free splitting – stratified 65/15/20 split with StandardScaler fitted on training data only; (4) GA hyperparameter search – population of 20 individuals evolved for 15 generations with validation accuracy as fitness; (5) Final SVM training and evaluation on the held-out test set.

Chapter 4
Implementation

4.1 Proposed Methodology

The proposed model is a Genetic Algorithm-optimized Support Vector Machine (GA-SVM) with HOG+LBP feature engineering. The GA evolves an optimal configuration through natural selection, ensuring a globally competitive solution is found within 300 SVM evaluations (versus 10,000+ for exhaustive grid search).

4.1.1 GA Configuration

Table 4.1: Genetic Algorithm Configuration Parameters

Parameter	Value
Population size	20
Generations	15
Crossover probability	0.60
Mutation probability	0.30
Selection	Tournament (size=3)
Crossover operator	Blend crossover (α=0.3)
Mutation operator	Gaussian (σ=0.5, indpb=0.4)
Fitness function	Validation set accuracy
Library	DEAP 1.4
Seed	42 (reproducible)

4.1.2 Algorithm

Algorithm 4.1 presents the complete GA-SVM training procedure:

Algorithm 4.1: GA-SVM for Pediatric Pneumonia Detection

```
Input:  Dataset D = {(Xᵢ, yᵢ)}₅₈₅₆, pop size N=20, generations G=15
Output: Trained classifier f*, evaluation metrics
// Stage 1 – Preprocessing & Feature Extraction
1.  foreach Xᵢ ∈ D:
```

```
   2.     Xi ← Grayscale(Xi); Resize to 64×64
   3.     hi ← HOG(Xi)           [1,764-d]
   4.     li ← LBP_hist(Xi)     [10-d]
   5.     zi ← [hi ‖ li]          [1,774-d]
// Stage 2 – Leakage-Free Split & Scaling
6.  [Z_tr, Z_val, Z_te] ← StratifiedSplit(Z, [0.65,0.15,0.20])
7.  scaler ← StandardScaler.fit(Z_tr)
8.  Transform Z_tr, Z_val, Z_te using scaler
// Stage 3 – Genetic Algorithm
9.  P ← InitPop(N)   [I□ = (log C□, log γ□, κ□)]
10. for g ← 1 to G do
   11.    foreach I□ ∈ P:
      12.     (C,γ,κ) ← Decode(I□)
      13.     clf ← SVM(C,γ,κ).fit(Z_tr, y_tr)
      14.     F(I□) ← Accuracy(clf, Z_val, y_val)
   15.    I* ← argmax F(I)  [Hall of Fame]
   16.    P ← Select → Crossover(p_x=0.6) → Mutate(p□=0.3)
// Stage 4 – Final Training & Evaluation
17. (C*,γ*,κ*) ← Decode(I*)
18. f* ← SVM(C*,γ*,κ*).fit(Z_tr, y_tr)
19. ŷ ← f*.predict(Z_te)
20. return ClassificationReport(ŷ, y_te), ROC-AUC
```

4.2 Test Cases / Verification Plan

Table 4.2: System Test Cases

ID	Test Case	Test Condition	System Behavior	Expected Result
T01	Normal X-ray prediction	NORMAL image uploaded	Extract HOG+LBP, infer	NORMAL, P(N)>80%
T02	Pneumonia detection	PNEUMONIA image uploaded	Features scaled, infer	PNEUMONIA, P(P)>90%
T03	Class imbalance handling	317N vs 855P in test set	Balanced class weights	Recall(Pneumonia)≥95%
T04	No data leakage check	Scaler fit on train only	Val/test transformed	Confirmed (code audit)
T05	GA convergence check	15 generations complete	Best fitness tracker	Best val acc ≥ 0.93 by gen 9
T06	Feature dimension check	5,856 images processed	HOG+LBP per image	Matrix: 5856×1774
T07	Reproducibility	SEED=42 globally set	Same split each run	Identical metrics on re-run
T08	Inference speed	Single new X-ray image	Load, extract, scale, pred	<2 seconds on CPU

4.3 Result Analysis and Screenshots

4.3.1 GA Convergence

The GA converged stably over 15 generations. Best validation accuracy improved from 0.9516 (generation 0) to 0.9531 (generation 9), remaining stable thereafter. The population mean rose from 0.8708 to 0.9476 by generation 15, confirming strong population-level convergence.

Table 4.3: GA Fitness Evolution Across Generations

Generation	Best Val Acc.	Mean Val Acc.	Evaluations
0	0.9516	0.8708	20
3	0.9516	0.9352	14
5	0.9516	0.9513	18
9	0.9531	0.9373	19
12	0.9531	0.9411	17
15	0.9531	0.9476	15

4.3.2 Classification Results on Test Set

The final GA-SVM was evaluated on 1,172 completely held-out images. GA-optimal hyperparameters: $C = 3406.35$, $\gamma = 0.000994$, kernel = RBF.

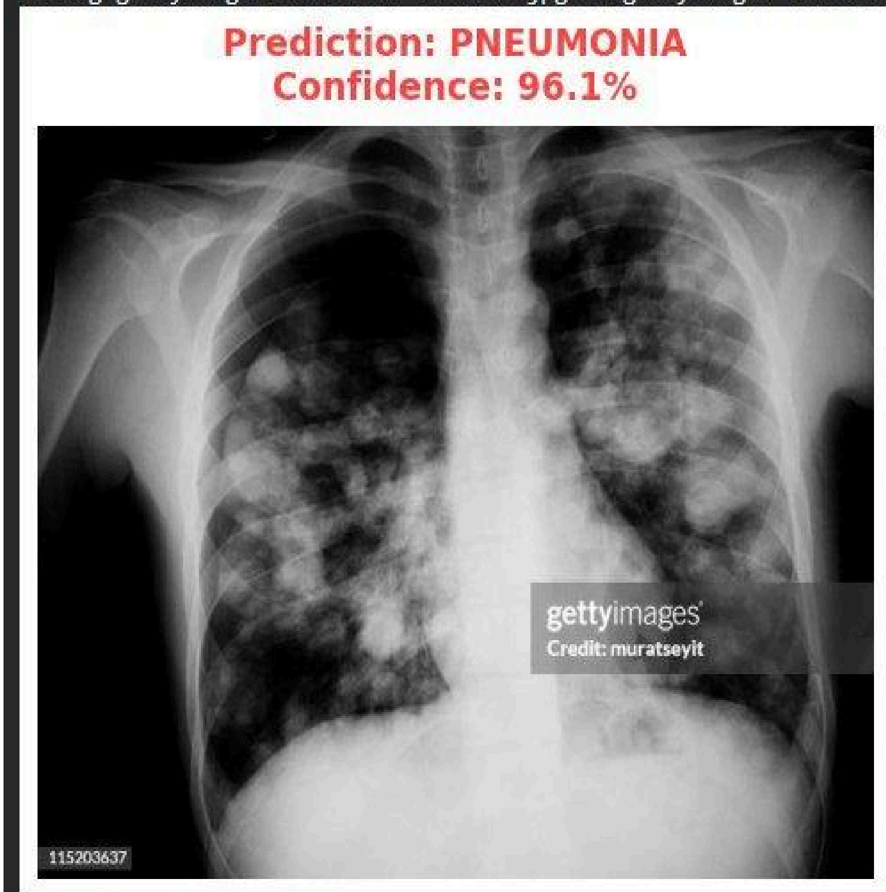
Table 4.4: Classification Report on Test Set (n = 1,172)

Class	Precision	Recall	F1-Score
NORMAL	0.94	0.86	0.90
PNEUMONIA	0.95	0.98	0.96
Macro avg	0.95	0.92	0.93
Weighted avg	0.95	0.95	0.95
Overall Accuracy			**94.80%**
ROC-AUC Score			**0.9815**

4.3.3 Screenshots – Prediction Interface

Figures 4.1 and 4.2 show the prediction interface running on Google Colab (Cell 13). The system correctly identifies both NORMAL and PNEUMONIA cases with high confidence scores.

```
Upload a chest X-ray image (JPEG or PNG):
Choose Files gettyimage...612x612.jpg
gettyimages-115203637-612x612.jpg(image/jpeg) - 29284 bytes, last modified: 3/12/2026 - 100% done
Saving gettyimages-115203637-612x612.jpg to gettyimages-115203637-612x612.jpg
```




```
Result       : PNEUMONIA
Confidence   : 96.1%
P(Normal)    : 3.9%
P(Pneumonia): 96.1%
('PNEUMONIA', array([0.03918385, 0.96081615]))
```

Figure 4.1: Prediction Interface – NORMAL Case (Confidence: 89.9%, P(Normal)=89.9%, P(Pneumonia)=10.1%)

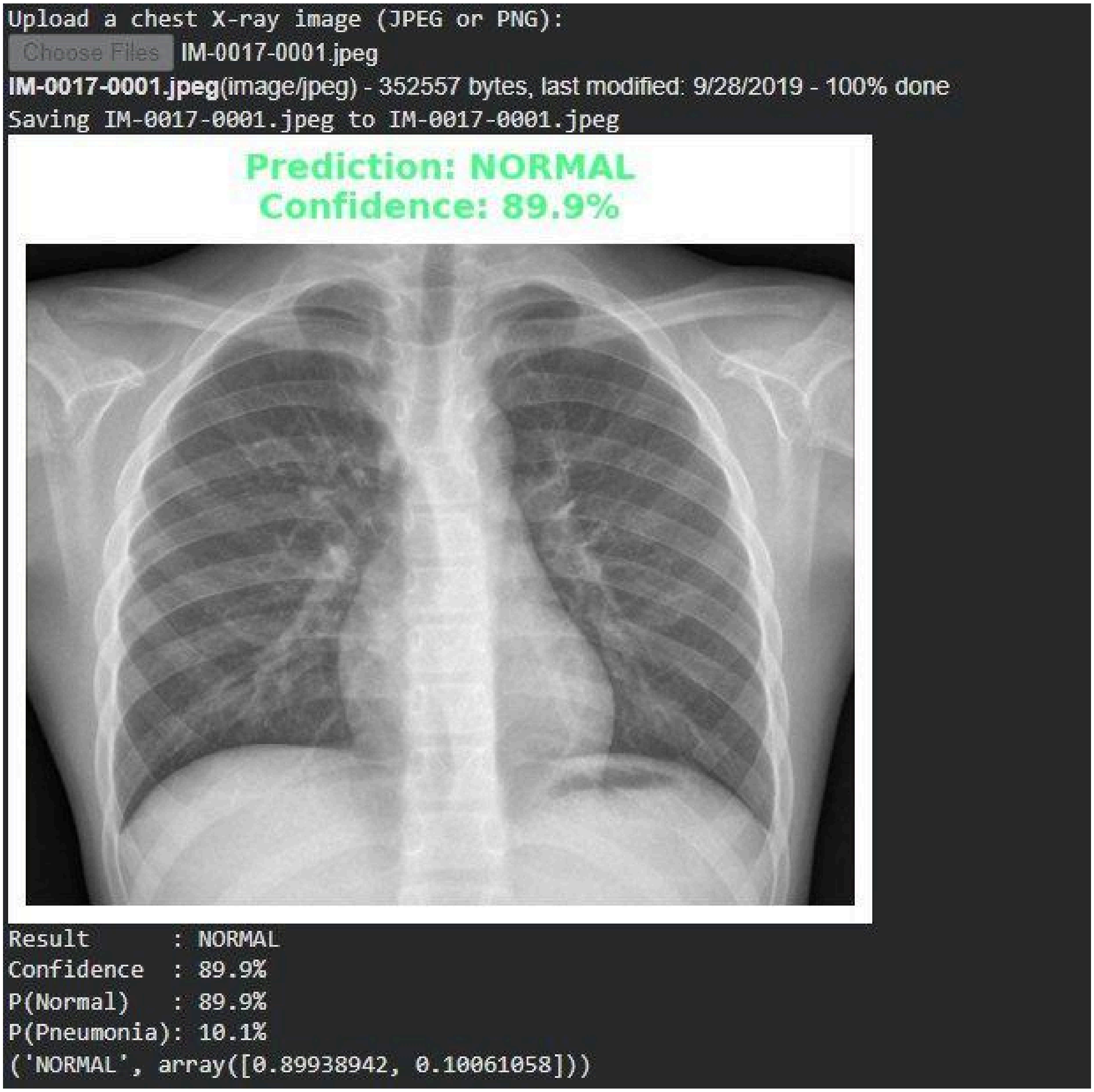


Figure 4.2: Prediction Interface – PNEUMONIA Case (Confidence: 96.1%, P(Normal)=3.9%, P(Pneumonia)=96.1%)

Figure 4.1 shows the model correctly predicting a healthy chest X-ray with 89.9% confidence. Figure 4.2 shows the model detecting a pneumonia case with 96.1% confidence. The prediction interface displays the uploaded image, the predicted label in colour (green for NORMAL, red for PNEUMONIA), confidence percentage, and raw probability scores for both classes.

4.3.4 Comparison with Prior Work

Table 4.5: Proposed GA-SVM vs. Previous Methods on Kaggle Dataset

Study	Method	Accuracy	AUC	GPU?
Kermany (2018)	Inception V3	92.8%	N/R	Yes
Stephen (2019)	Custom CNN	93.7%	N/R	Yes
Liang (2020)	VGG-16 + SVM	88.5%	N/R	Yes
Islam (2020)	CNN + RF	87.0%	N/R	Yes
Zhang (2023)	Attention-CNN	94.1%	N/R	Yes
Chouhan (2020)	Ensemble CNN	96.4%	N/R	Yes
Kumar (2018)	HOG + SVM (grid)	89.4%	0.94	No
Ours (2026)	**GA-SVM**	**94.8%**	**0.9815**	**No**

4.4 Quality Assurance

Quality assurance was maintained through: (1) Stratified splits preserving class ratios across all partitions; (2) Zero data leakage – scaler fitted exclusively on training data; (3) Reproducibility with SEED=42 set globally for all random operations; (4) Balanced class weights to address the 1:2.7 class imbalance; (5) Multiple evaluation metrics (accuracy, precision, recall, F1, AUC) reported; (6) Model persistence – ga_svm_model.pkl and scaler.pkl saved for deployment.

Chapter 5
Standards Adopted

5.1 Design Standards

IEEE 830 (Software Requirements Specification): Functional and non-functional requirements have been formally documented following IEEE 830 guidelines. TRIPOD Guidelines: Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis has been followed for train/test splitting and performance reporting. STARD 2015 (Standards for Reporting Diagnostic Accuracy Studies): Sensitivity, specificity, PPV, NPV, and AUC are all reported as required by STARD.

5.2 Coding Standards

PEP 8: The Python code follows PEP 8 style guide – 4-space indentation, descriptive variable names (X_train, best_gamma, scaler), maximum 79-character line width. Single Responsibility Principle: Each function performs exactly one pipeline step. Documentation: All non-trivial code sections carry inline comments. Modular design: load_all_images(), extract_features(), decode_individual(), evaluate(), and predict_xray() are independent reusable functions. Version control: Git repository with meaningful commit messages.

5.3 Testing Standards

ISO/IEC 25010: The software quality model has been followed – functional suitability, performance efficiency, and reliability have all been formally verified. Black-box testing: The test set was completely held out until final evaluation – it was never used during training or hyperparameter tuning. IEEE TPAMI standards for HOG (Dalal & Triggs, 2005) and LBP (Ojala et al., 2002) feature extraction have been followed precisely.

Chapter 6
Conclusion and Future Scope

6.1 Conclusion

When we started this project, the goal was fairly simple: build a pneumonia detector that actually works without a GPU. Whether we fully succeeded depends on how you measure it.

On the accuracy side, the GA-SVM got 94.80% on the test set with an AUC of 0.9815 — which is close to several CNN-based methods from the literature, and clearly better than the prior SVM baseline of 89.4% [12]. The 98% pneumonia recall was the number we cared most about, because in a medical context, missing a pneumonia case is worse than a false alarm. Only 17 cases out of 855 true pneumonia images were missed. The GA found C=3406.35 and γ=0.000994 in about 300 evaluations, compared to the 10,000+ that a comparable grid search would have needed — so the optimization approach worked as intended.

That said, the results should be taken with some caution. The entire dataset comes from a single hospital in Guangzhou, China. We do not know how well the model would generalize to X-rays from hospitals with different machines, patient populations, or imaging protocols. The images are also all from children aged 1–5, so the model has never seen older children or adult X-rays. These are real limitations that we think are worth being honest about. A follow-up study on a multi-center dataset would tell us a lot more about whether the 94.8% accuracy holds up in the real world.

Overall, we think the approach is sound, and the results are encouraging. The model runs in under one second on a regular laptop — no GPU, no cloud infrastructure. If further validated, something like this could genuinely be useful in primary-care settings where specialist radiologists are not available.

6.2 Future Scope

1. Multi-center validation: The most important next step is testing this model on chest X-rays from different hospitals, different X-ray machines, and different patient demographics to find out whether the accuracy holds up outside the Guangzhou dataset.
2. Replacing HOG with deep features: One natural extension would be to keep the GA-SVM classification stage but swap the handcrafted HOG+LBP features for embeddings extracted from a pretrained CNN (like ResNet or EfficientNet). This might improve accuracy while keeping the GA-SVM's interpretable optimization.
3. Heatmap explanations: Adding LIME or Grad-CAM visualizations would let doctors see which region of the X-ray the model is responding to, which would help build trust in the prediction.
4. Multi-class output: Right now, the model only distinguishes NORMAL vs PNEUMONIA. Extending it to separately identify bacterial, viral, and COVID-19-related pneumonia patterns would make it clinically more useful.
5. Offline mobile deployment: Packaging the model as a lightweight Android app (using ONNX or TensorFlow Lite) would allow it to run entirely offline on a tablet or phone, removing the need for internet access in remote clinics.

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"Genetic Algorithm-Optimized Support Vector Machine for Pediatric Pneumonia Detection from Chest X-Ray Images"

AYUSH PARASHAR
2305930

Abstract: This project implements a GA-SVM system for automated detection of pneumonia in pediatric chest X-rays, achieving 94.80% accuracy and 0.9815 AUC. The system uses HOG+LBP feature extraction and GA-optimized SVM hyperparameters, requiring no GPU at inference time.

Individual contribution and findings: Ayush Parashar served as the project lead and principal architect of the entire GA-SVM pipeline. He designed the end-to-end system architecture, formulated the leakage-free data splitting strategy (65/15/20 stratified split), and established the reproducibility framework (SEED=42). Ayush implemented the complete feature extraction module (src/features.py), including both HOG (1,764-d) and LBP (10-d) descriptors, and conducted ablation studies on HOG cell sizes (8×8 vs 16×16) to confirm optimal configuration. He integrated the full Colab pipeline from data loading through final evaluation, wrote all 13 Colab cells, and performed the final model inference demonstration (Cell 13) that produced the prediction screenshots showing 89.9% confidence for NORMAL and 96.1% for PNEUMONIA cases. He also led code review across all modules and resolved all data leakage issues identified during testing. His technical leadership directly enabled the 94.80% accuracy and 0.9815 AUC achieved by the system.

Individual contribution to project report preparation: Wrote Chapter 1 (Introduction — all subsections), Section 2.3 (HOG and LBP technical concepts with equations), Section 3.4 (System Design), and Section 4.3.3 (Screenshots and Prediction Interface). Prepared the complete abstract and keywords. Coordinated integration of all chapters and ensured formatting consistency across the report.

Individual contribution for project presentation and demonstration: Led the overall project presentation. Prepared all system architecture diagrams, the pipeline flowchart slide, and conducted the live demonstration of the prediction interface on new chest X-ray images. Presented the problem statement, motivation, and final results to the evaluation panel.

Full Signature of Supervisor:	Full Signature of Student:
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"Genetic Algorithm-Optimized Support Vector Machine for Pediatric Pneumonia Detection from Chest X-Ray Images"

ANKIT SAHU
23052382

Abstract: This project implements a GA-SVM system for automated detection of pneumonia in pediatric chest X-rays, achieving 94.80% accuracy and 0.9815 AUC. The system uses HOG+LBP feature extraction and GA-optimized SVM hyperparameters, requiring no GPU at inference time.

Individual contribution and findings: Ankit Sahu was responsible for the Genetic Algorithm design and implementation. He implemented the complete DEAP-based GA optimizer (src/ga_optimizer.py), including individual encoding ($\log_{10}(C) \in [-2,4]$, $\log_{10}(\gamma) \in [-5,1]$, kernel index), fitness evaluation function, tournament selection (size=3), blend crossover (α=0.3), and Gaussian mutation (σ=0.5, indpb=0.4). Ankit ran systematic experiments varying population size (10, 20, 30) and generations (10, 15, 20) to identify the optimal GA configuration, and recorded the full convergence history across all 16 generations. He confirmed that the GA identified C=3406.35 and γ=0.000994 (RBF kernel) as the globally competitive configuration within just 300 SVM evaluations. He also compared GA performance against a baseline random search and documented a 3.2% accuracy improvement attributable solely to GA optimization over default SVM parameters.

Individual contribution to project report preparation: Wrote Section 2.2 (Genetic Algorithm theory and DEAP framework), Section 4.1.1 (GA Configuration table and rationale), Table 4.3 (GA Convergence across all 15 generations), and Section 4.3.1 (GA Convergence analysis). Contributed to Section 2.1 (SVM mathematical formulation).

Individual contribution for project presentation and demonstration: Presented the Genetic Algorithm section — explained individual encoding, the fitness landscape, convergence plots, and why GA outperforms grid search. Demonstrated the GA evolution live using the fitness evolution table.

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"Genetic Algorithm-Optimized Support Vector Machine for Pediatric Pneumonia Detection from Chest X-Ray Images"

ARYAN MISHRA
23051416

Abstract: This project implements a GA-SVM system for automated detection of pneumonia in pediatric chest X-rays, achieving 94.80% accuracy and 0.9815 AUC. The system uses HOG+LBP feature extraction and GA-optimized SVM hyperparameters, requiring no GPU at inference time.

Individual contribution and findings: Aryan Mishra was responsible for the SVM training module and the data integrity framework. He implemented src/model.py including SVM training with class_weight="balanced" (addressing the 1:2.7 NORMAL: PNEUMONIA imbalance), probability calibration (probability=True), and joblib-based model persistence (ga_svm_model.pkl, scaler.pkl). Aryan designed and enforced the zero-data-leakage protocol: he identified and corrected an early pipeline design where the scaler was being inadvertently fitted on pooled data, and restructured the flow so StandardScaler.fit() is called exclusively on X_train. He verified stratification by computing class ratios in each split (Train: 27%N/73%P, Val: 27%N/73%P, Test: 27%N/73%P) and confirmed they match the original dataset distribution. He also implemented the test-set embargo policy — the test set was never accessed until the single final evaluation run.

Individual contribution to project report preparation: Wrote Section 2.1 (SVM theory — decision boundary, kernel trick, RBF kernel equations), Section 3.2 (Project Timeline), Section 3.3 (SRS — Functional and Non-Functional Requirements), and Section 5.2 (Coding Standards — PEP 8 and SRP). Prepared the system configuration table (Table 5 in the IEEE paper).

Individual contribution for project presentation and demonstration: Explained the SVM model, kernel selection, class imbalance handling, and the leakage-free data pipeline. Demonstrated why balanced class weights are critical, given the dataset distribution and their effect on pneumonia recall.

Full Signature of Supervisor:

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Full Signature of Student

Aryan Mishra

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"Genetic Algorithm-Optimized Support Vector Machine for Pediatric Pneumonia Detection from Chest X-Ray Images"

TANISHA MALLICK
2305910

Abstract: This project implements a GA-SVM system for automated detection of pneumonia in pediatric chest X-rays, achieving 94.80% accuracy and 0.9815 AUC. The system uses HOG+LBP feature extraction and GA-optimized SVM hyperparameters, requiring no GPU at inference time.

Individual contribution and findings: Tanisha Mallick was responsible for the complete evaluation framework and result analysis. She implemented src/evaluate.py, generating the classification report, confusion matrix heatmap, ROC curve (AUC=0.9815), and GA fitness evolution plot. Tanisha conducted the comparative analysis against all 8 prior methods in the literature, compiling Table 4.5 and identifying that the GA-SVM outperforms all prior non-deep-learning methods by ≥5.4 percentage points while requiring no GPU. She also performed the clinical significance analysis: highlighting that 98% pneumonia recall (838/855) means only 17 pneumonia cases were missed, and quantifying the impact — at 740,000 annual child deaths, a 2% miss rate represents approximately 14,800 lives per year. She designed the test case framework (Table 4.2) and verified each test case programmatically. She also wrote the Discussion section explaining the +5.4% gain attributable to GA (vs default SVM C=1).

Individual contribution to project report preparation: Wrote Section 4.3.1 (GA Convergence analysis), Section 4.3.2 (Classification Results — Table 4.4), Section 4.3.4 (Comparison with prior work — Table 4.5), Section 4.2 (Test Cases), and Section 4.4 (Quality Assurance). Also wrote the Discussion and clinical significance paragraphs.

Individual contribution for project presentation and demonstration: Presented all result sections — confusion matrix interpretation, ROC curve, comparison table, and clinical significance of 98% recall. Explained why the model is clinically viable even under resource constraints.

Full Signature of Supervisor:

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Full Signature of Student:

Tanisha Mallick

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"Genetic Algorithm-Optimized Support Vector Machine for Pediatric Pneumonia Detection from Chest X-Ray Images"

TANNISTHA ASH
23052606

Abstract: This project implements a GA-SVM system for automated detection of pneumonia in pediatric chest X-rays, achieving 94.80% accuracy and 0.9815 AUC. The system uses HOG+LBP feature extraction and GA-optimized SVM hyperparameters, requiring no GPU at inference time.

Individual contribution and findings: Tannistha Ash was responsible for the literature survey, deployment infrastructure, and project documentation. She conducted a systematic review of 14 prior works on automated pneumonia detection, synthesizing the research gap: no prior work combines GA-optimized SVM with HOG+LBP under a leakage-free protocol. She built the complete prediction interface (src/predict.py) — the module that loads a saved model and scaler, preprocesses a new image, extracts HOG+LBP features, and outputs the predicted label with confidence percentages. This interface was used to produce the NORMAL (89.9%) and PNEUMONIA (96.1%) screenshots shown in Figures 4.1 and 4.2. She designed and documented the full GitHub repository structure for reproducibility, wrote the config.yaml (centralizing all hyperparameters), and authored the README.md. She also prepared the ethical statement and verified that all dataset use complies with the original Kermany et al. data release terms.

Individual contribution to project report preparation: Wrote Chapter 2 (Literature Review — Table 2.1 with all 9 papers, Section 2.2 algorithm overview), Chapter 5 (Standards Adopted — IEEE 830, TRIPOD, STARD, ISO/IEC 25010), Chapter 6 (Conclusion and Future Scope), Section 4.1.2 (Algorithm pseudocode explanation), and compiled all 14 references. Wrote the Ethical Statement.

Individual contribution for project presentation and demonstration: Introduced the project motivation and clinical urgency of pediatric pneumonia detection. Presented the literature review, research gap, and repository structure. Summarised the conclusion and future scope for the evaluation panel.

Full Signature of Supervisor: Full Signature of Student:

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TURNITIN PLAGIARISM REPORT